

What `landscape_spectrum` plots

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Setup

Let $A \in \mathbb{R}^{n \times n}$ be the interior block of the Q_1 stiffness matrix on the $N \times N$ grid ($N = 64$, so $n = (N - 2)^2 = 3844$), $b = Mf$ the consistent load, and $u^\star = A^{-1}b$ computed by a direct solve in `float64`. Adam is run in `float32` on

$$L(u) = \frac{1}{2} \|Au - b\|_2^2, \quad u_0 = 0,$$

producing iterates u_k . The quantity the figure decomposes is the *error*

$$e_k = u_k - u^\star \in \mathbb{R}^n, \quad \text{so} \quad e_0 = -u^\star.$$

The basis

On a uniform grid $A = K \otimes \tilde{M} + \tilde{M} \otimes K$ with $K, \tilde{M}$ the 1D stiffness and mass matrices, and the 1D sine vectors diagonalise both. Hence the 2D sine modes

$$(v_{pq})_{ij} \propto \sin\left(\frac{\pi p i}{N-1}\right) \sin\left(\frac{\pi q j}{N-1}\right), \quad p, q = 1, \dots, N-2,$$

normalised to $\|v_{pq}\|_2 = 1$, are *exact* eigenvectors of A ,

$$A v_{pq} = \lambda_{pq} v_{pq}, \quad \lambda_{pq} = k_p m_q + m_p k_q,$$

with $\theta_p = \pi p h$, $h = 1/(N-1)$, $k_p = h^{-1}(2 - 2 \cos \theta_p)$ and $m_p = \frac{h}{6}(4 + 2 \cos \theta_p)$. They are orthonormal, so

$$e_k = \sum_{p,q} c_{pq}(k) v_{pq}, \quad c_{pq}(k) = \langle e_k, v_{pq} \rangle, \quad \|e_k\|_2^2 = \sum_{p,q} |c_{pq}(k)|^2.$$

The $c_{pq}(k)$ are obtained from one orthonormal 2D DST-I of e_k ; no eigensolver is used. The eigenvector property is verified numerically at run time (relative residual $\sim 10^{-13}$).

The plotted quantity

The same construction is applied to the gradient-descent iterates, giving the companion figure `landscape_spectrum_gd`.

Write $\hat{c}_{pq}(k)$ for the root-mean-square of $|c_{pq}(k)|$ over the 25 modes nearest in λ (a denoising window along the mode axis only, normalised by the number of samples actually inside it; the λ axis is not binned). The colour is

$$\boxed{\log_{10} \frac{\hat{c}_{pq}(k)}{\|e_0\|_2}} \quad \text{clipped to } [-5, 0],$$

plotted against iteration k (horizontal, log) and eigenvalue λ_{pq} (vertical, log). Each cell is one recorded iteration $\times$ one mode.

Why the normalisation is global, not per mode. The natural choice $|c_{pq}(k)|/|c_{pq}(0)|$ is undefined here: with $u_0 = 0$ the initial error is $e_0 = -u^*$, and u^* inherits the K^2 frequencies of the source, so **only 16 of the 3844 coefficients $c_{pq}(0)$ are nonzero**. Dividing by $\|e_0\|_2$ instead makes every cell comparable and keeps Parseval available: the squared cell values in a column sum to $\|e_k\|_2^2/\|e_0\|_2^2$, the squared relative error.

Reading it, and one trap. Bright = this mode carries error comparable to the *total* initial error; black = at or below 10^{-5} of it. The colour is a *per-mode* amplitude, and the bands hold very different numbers of modes: 3511 modes have $\lambda \geq 1$ and only 26 have $\lambda < 0.1$. **A broad dim band can therefore carry far more error than a narrow bright one**, and the heatmap alone does not support quantitative statements about where the error lives. The companion figure `landscape_bands` plots the un-diluted quantity, the fraction of $\|e_k\|_2^2$ in each band, and that is the one to quote.

What the two optimizers actually do. They behave oppositely, and only the gradient-descent panel matches the classical intuition.

- **Gradient descent** is the textbook smoother: everything above $\lambda \approx 0.1$ is annihilated within the first iteration, and 100% of the squared error sits in $\lambda < 0.1$ from iteration 1 through 10^5 , essentially undiminished. It ends at 46.7% relative error.
- **Adam** does the reverse. The smooth band dominates until $k \approx 10^3$, then crosses over sharply around $k \approx 3\text{--}7 \times 10^3$, and by $k \approx 10^4$ the modes with $\lambda \geq 1$ carry 100% of the error, with the smooth coefficients down at $\sim 5 \times 10^{-5}$. Adam *solves* the smooth content and is left rattling in the stiff modes. This is consistent with its per-coordinate normalisation: it advances every coordinate by $\sim \text{lr}$ regardless of curvature, so it has no difficulty with flat directions, and no way to settle in steep ones.

So the ill-conditioning of $\frac{1}{2}\|Au - b\|^2$ defeats both, but for different reasons: GD cannot move along the flat directions, while Adam cannot stop moving along the steep ones.

Two rendering notes

- **Cell widths are unequal by construction.** Iterations are recorded log-spaced but must be integers, so the first ones are 1, 2, 3, $\dots$; on a log axis the step $1 \rightarrow 2$ spans 0.30 decades while $10^4 \rightarrow 10^5$ spans 1. The wide left-hand cells are single iterations, and the strong colour changes between them are real: the amplitudes genuinely swing by an order of magnitude per step during the first few iterations.

- Cell edges are the geometric midpoints between samples, with the first edge pinned at $k = 1$. (With `shading="nearest"` the first cell is centred on $k = 1$ and extends half a log-spacing to its left, which made the axis display iterations $< 10^0$.)